

	COMPASS Field Sampling Protocol Site: Synoptic Sites	
Created: 23 June 2022 Updated:	Created by: S. Wilson	Version: 1.0
Redox Probes: Field Use & Lab Care		

I. Objective:

To measure redox profiles in soil. This protocol describes the equipment, use of probes in the field, and lab care/storage.

II. Contents:

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III. Experimental Design:

Each Synoptic site will have a set of redox probes (5), a reference probe (1), and a handheld sensor readout. These will be used to characterize profiles of redox in soils in the upland, transition and wetland plots.

IV. Redox Probe Overview:

The redox probes, supplied by Paleo Terra & XXX, will be used to measure soil redox profiles at synoptic sites.

Each redox kit contains five probes (provided by SERC), one Calomel (Ag/AgCl) reference probe (provided by SERC), and a handheld readout (provided by the respective location CB or LE). The probes have platinum rings at one or two depth that serve as the “sensor” for redox potential. The voltage difference between the sensor probe and the reference probe results in the measurement.

Note: Redox measurements can be very variable and take time to stabilize.

V. Personal Protective Equipment:

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Close-toed shoes and long pants are required at all times when working at any COMPASS site. Nitrile grip work gloves can be used and are available if needed. Protected eye and ear wear are available.

VI. Materials List

A. Materials

- 50mL Falcon Tube (to store reference probe)
- Potassium chloride (KCl)
- DI Water
- Paper towels/kim wipes

B. Probe Installation & Measurement

- Redox probes
- Reference probe
- Redox handheld readout
- Driveway stake (or similar for guide hole)
- Rags / Paper towels
- 3M KCl
- Saturated KCl
- KCl powder
- Markers
- Tape
- Ruler
- Datasheet
- Pencil/Pen

VII. Probe Storage, Care, and Prep:

Redox probe storage, care and preparation can be done in the lab and the probes should be cleaned and checked before and after use.

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A. Reference Probe

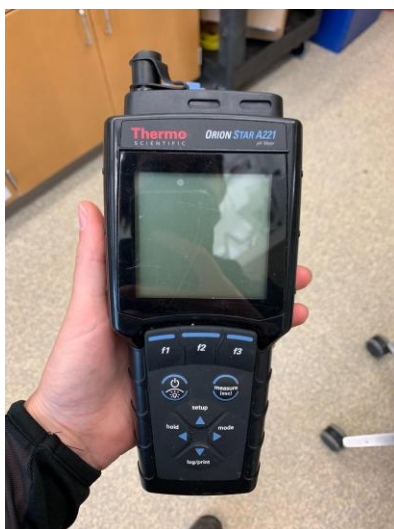
1. The reference probe is filled with saturated KCl. Crystals of KCl are placed in the bottom of the reference probe as well.
2. The reference probe is stored in 3M KCL in a 50mL falcon tube (wrap the cap with parafilm to prevent leaks)
 - a) Refresh the KCl in the falcon tube after each sampling if it is dirty

B. Redox Probes

1. Should be kept in a bin to prevent wear and tear
2. Wipe the redox probes clean after use with a rag or paper towel
3. Check the ends of the wires for damage

C. Handheld Readout

1. SERC is using a Thermo Scientific Orion Star A221 pH Meter
 - This runs on batteries so you should keep extra on hand in the redox bin
 - The pH meter comes with a BNC connector to attach the meter to the redox electrodes and a pin connector to connect the reference electrode (see Fig. below)



D. Calibration / Checking Function of Probes

1. See attached supplemental file for Quinone solution calibration

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VIII. Field Use & Measurement:

A. Probe Field Use

1. Select an area to take your measurements. These are replicates so 10 cm spacing is sufficient.
 - The Synoptic CB team is making these in each zone near to the TEROS sensors
2. Use the driveway stake to create guide hole to 3-5 cm above the desired depth
3. Push the redox probe into the hole to the desired depth (i.e. 3-5 cm past the depth of the guide hole).
 - It is VERY important not to push or pull on the wires! Put your hand on the hard plastic part of the probe and push down into the ground.
4. Repeat with as many probes as you plan to measure from. Redox is highly variable so we recommend using at least 5 replicate probes.
5. Create a hole for the reference electrode (e.g. with a syringe plunger).
6. Take the reference electrode out of the falcon tube with KCl then push the reference probe into the ground as well.
7. Let the probes sit in the ground/soil to equilibrate for 5-10 mins. By the time all the probes are inserted enough time will have elapsed to read the first probe.
8. Connect the reference probe to the handheld pH readout (push reference wire into small circular hole on top of handheld readout where it says "REF")
9. Turn on the handheld readout (press power button)
10. Connect the sensor wire to the metal connector and attach the clamp to the first reference probe
11. Press the mode? button until the units are mV
12. If not already measuring (noticeable because the value on the display is fixed), push the measure button on the handheld readout
13. Allow the numbers to settle until the display fixes a number or until a fixed amount of time has elapsed (e.g. 2 minutes) and record your reading
 - Readings can be y variable and often do not completely settle
14. Repeat steps 9-11 for each probe by connecting the clamp to each of the probes one by one
15. After taking measurements you can push the probe to the next desired depth or you can remove it from the ground carefully - DO NOT PULL ON THE WIRES!

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IX. Monitoring:

Redox measurements should be taken monthly or bimonthly at synoptic sites until these measurements can be automated.

A. Readings

Ideal/normal readings will depend on the site and depth of the measurement. However, as the system becomes more anoxic/reducing the redox measurement should get lower / more negative.

B. Field Datasheet

Creating a field datasheet is recommended to keep track measurements in the field. The synoptic CB field team's sheet has been pasted as an example at the end of this protocol in corresponding documentation.

X. Data Analysis:

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A. Readings

IMPORTANT!: With this Redox set we need to **add 202 mV** to the measurements to adjust from using a saturated KCl reference electrode to the standard hydrogen electrode (SHE), based on values found in the literature.

The table below shows the readings for the thermodynamic sequence at 25 C and pH 7:

Reaction	Eh (mV)
O ₂ Reduction	812
NO ₃ ⁻ Reduction	747
Mn ⁴⁺ to Mn ²⁺	526
Fe ³⁺ to Fe ²⁺	-47
SO ₄ ²⁻ to H ₂ S	-221
CO ₂ to CH ₄	-244

B. Effect of pH

When interpreting redox potentials recognize that pH affects redox. There are corrections for normalizing reading for pH but they are approximate so the pros and cons should be discussed before using a correction.

XI. Troubleshooting:

Any issues with the redox probes should be reported to the SERC Biogeochem lab.

A. Knowing How it Works

Review this protocol and associated documentation to learn more about how the redox probes are measuring redox potential.

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B. Unusual Readings

If you observe a reading that seems way out of range, be sure that you have connected your readout to the correct wire (readout wire not the ground). Keep in mind that wide variation and soil patchiness can lead to highly variable readouts, but you will likely see a pattern with depth and zone (Wetland, Transition, Upland).

Power Issues

Check handheld readout batteries

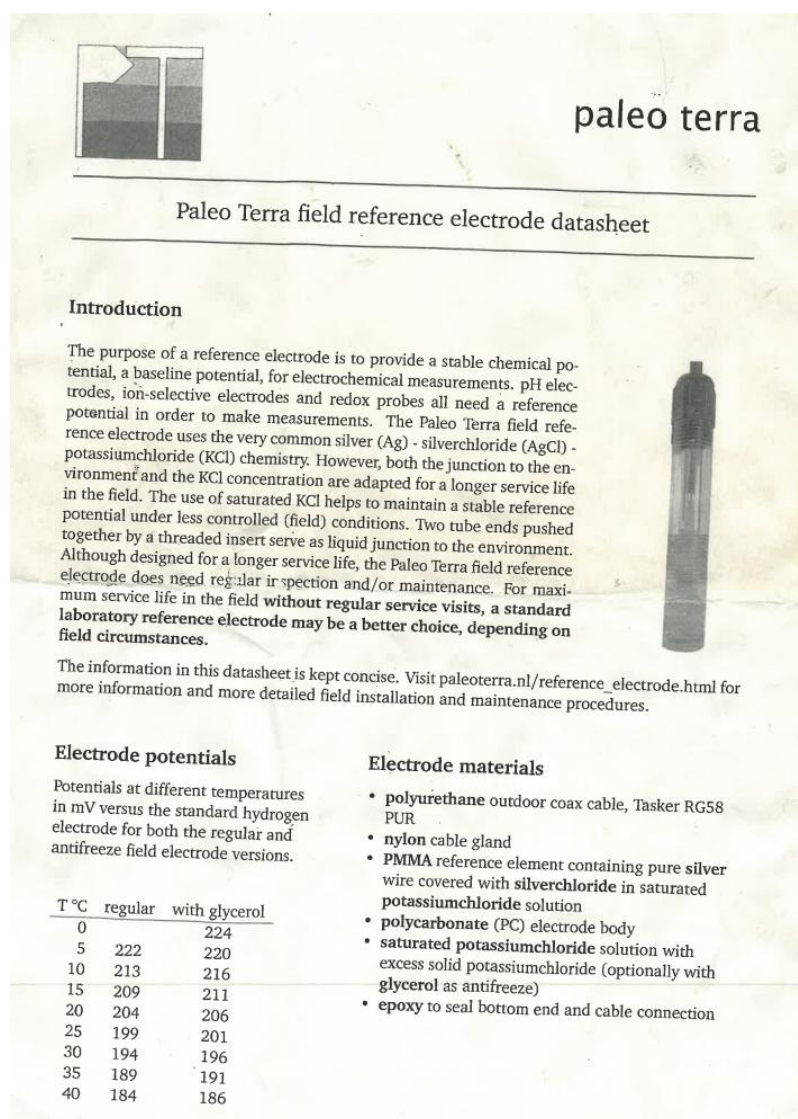
XII. Corresponding documentation:

A. Sampling design:

B. Manuals

1. Redox reference electrode (One used in this protocol)

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paleo terra

Paleo Terra field reference electrode datasheet

Introduction

The purpose of a reference electrode is to provide a stable chemical potential, a baseline potential, for electrochemical measurements. pH electrodes, ion-selective electrodes and redox probes all need a reference potential in order to make measurements. The Paleo Terra field reference electrode uses the very common silver (Ag) - silverchloride (AgCl) - potassiumchloride (KCl) chemistry. However, both the junction to the environment and the KCl concentration are adapted for a longer service life in the field. The use of saturated KCl helps to maintain a stable reference potential under less controlled (field) conditions. Two tube ends pushed together by a threaded insert serve as liquid junction to the environment. Although designed for a longer service life, the Paleo Terra field reference electrode does need regular inspection and/or maintenance. For maximum service life in the field without regular service visits, a standard laboratory reference electrode may be a better choice, depending on field circumstances.

The information in this datasheet is kept concise. Visit paleoterra.nl/reference_electrode.html for more information and more detailed field installation and maintenance procedures.

Electrode potentials

Potentials at different temperatures in mV versus the standard hydrogen electrode for both the regular and antifreeze field electrode versions.

T °C	regular	with glycerol
0		224
5	222	220
10	213	216
15	209	211
20	204	206
25	199	201
30	194	196
35	189	191
40	184	186

Electrode materials

- polyurethane outdoor coax cable, Tasker RG58 PUR
- nylon cable gland
- PMMA reference element containing pure silver wire covered with silverchloride in saturated potassiumchloride solution
- polycarbonate (PC) electrode body
- saturated potassiumchloride solution with excess solid potassiumchloride (optionally with glycerol as antifreeze)
- epoxy to seal bottom end and cable connection



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Field installation

- One reference electrode can serve multiple redox probes, as long as an electrically conductive path exists between redox probes and the reference electrode. Moist and wet soil conduct well, as most natural waters. Dryer soil may not conduct sufficiently; plastic liners or containers separating experiments do not conduct.
- Before installation, adjust the liquid junction to your needs. The liquid junction is located where threading can be seen inside the clear tube (below the internal reference element, above the excess KCl). Remove any protective film if still present. Two pieces of tube are connected with a threaded insert. The tube ends are very smooth and when pushed together they form the liquid junction. For a reference electrode to work, some transport of KCl through the liquid junction to the outside environment must be allowed. Therefore, the tube ends must be loosened a little. Easier flow of KCl minimizes the liquid junction potential, but will release more KCl to the environment resulting in a shorter maintenance interval. In most applications, screwing the tube ends together without applying much force results in a good balance between easy flow of KCl and little KCl loss. Check by adjusting and then leaving the reference electrode overnight. By the next day, a visible stain of KCl must have formed on the outside, but the extent of the stain should be limited.
- Preferably install in clear water, e.g. into the groundwater via a dipwell or directly into a surface water.
- Do not install in wet clay or a clay suspension: a large error (junction potential) may arise due to the "suspension effect."
- Install at a depth where temperatures don't drop below -5°C (-20°C for electrodes with antifreeze)

Regular checks

- Reference electrodes installed in a place that may dry out periodically should be checked very regularly, e.g. every week. At other places monthly checks to perhaps once every three months are recommended. If installed into permanent groundwater or a surface water and with plenty excess KCl present checking twice a year may be appropriate. 1 gram KCl lasts for 3-6 months.
- Clean the electrode with a piece of cloth to remove dirt, build-up of micro-organisms, etc.
- The filling solution should be a clear and colourless liquid.
- Excess KCl, a white solid, should be present in the filling solution.
- Check for the absence of air bubbles. A few small bubbles are acceptable.
- The liquid junction should be clean to allow the exchange of ions between the internal solution and the outer environment.
- Check the cable for any damage.
- If possible, check the reference electrode voltage against a fresh reference electrode. A difference of a few mV (up to perhaps 10 mV) is acceptable.

Maintenance

Regular maintenance may include refilling of the internal KCl solution, replenishing the solid KCl and cleaning the threading of the liquid junction. Proceed as follows to replenish or refill the KCl solution and/or add solid KCl:

- prepare a saturated KCl solution: add 40 gram solid KCl to 100 ml distilled water; stir or shake until almost all KCl has dissolved; at 20°C, 34 gram KCl can dissolve in 100 ml water, and 37,4 gram at 30°C
- clean the outside of the electrode body
- unscrew the top cable gland with reference element from the clear electrode body; use a 15mm spanner to hold the cable gland; be careful not to open the cable gland itself
- clean the outside of the reference element; if possible, keep the element wet during the remainder of this procedure
- if the KCl solution may be polluted (discoloration of the solution and/or presence of another solid than white KCl): dispose of the solution and rinse with fresh KCl solution
- if the threading at the liquid junction is polluted: unscrew, clean and re-assemble
- if solid KCl must be replenished: (re)fill with saturated KCl solution until the threading at the liquid junction is fully submerged, then add solid KCl; 1 gram lasts 3-6 months
- top up with saturated KCl solution
- place the reference element into the solution and screw into place; some KCl solution will overflow; fasten tight, but not overly tight

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2. Procedure by Karen McKee & Peter Mueller **NOTE: THIS PROCEDURE USES A DIFFERENT SET OF PROBES AND REFERENCE ELECTRODE - This is only for reference and the electrode check solution readings will be different for our probes.**

PROCEDURE FOR MEASURING SOIL REDOX POTENTIAL

Karen L. McKee, Peter Mueller

The original author of this document is Karen L. McKee. The following is a modified version for the general procedure of measuring soil redox potentials using material and instruments of the biogeochemistry lab.

Principles

Redox electrodes are measuring “electron pressure”. Reactions in the soil involve electron transfers, and electrons build up when they are not being accepted. You are measuring this buildup or potential. The Nernst Equation describes these relationships. Essentially, you are measuring the oxidizing relative to reducing reactions in the soil.

An inert metal (Pt or gold) is used in construction of electrodes since it does not participate in the reactions producing the electrons. The Pt electrode is a half cell which is connected to another half cell to complete the circuit. You are measuring the difference between the two. The second half cell is the reference electrode, which has a known potential. The reading you see on the meter is the potential of the whole cell. You want to know the potential of the half cell (Pt). Thus, you must subtract the known potential of the reference from the whole cell measurement to calculate Eh (although in actual practice you add the two numbers). Also keep in mind that you are measuring a “mixed potential” in soils. Thus, it is impossible to relate Eh of natural systems to the EMF of chemical systems (as defined by the Nernst Equation). It is also impossible to “correct” for pH because you must know which reactions are dominating the system in order to calculate the appropriate correction factor (based on the electrons being transferred in the reaction). It is also usually unnecessary to correct for temperature since Eh changes only 1 mV for each degree C change in temperature (this is far smaller than the variance due to other factors—usually 40-50 mV). Oxygen disappears at +300 mV. Typical redox values (corrected) for wetland soils vary from +400 to -250 mV.

Flooded soils go through a predictable sequence of reductions depending upon their potential energy: O_2 , NO_3^- , Mn^{4+} , Fe^{3+} , SO_4^{2-} and CO_2 are sequentially reduced to H_2O , N_2 , Mn^{2+} , Fe^{2+} , S^{2-} , and CH_4 . The former are called oxidants since they receive electrons during these reactions. These reductions occur at a specific Eh (generally). These are biochemical reactions mediated by microorganisms. An energy source is required to carry out these reductions: many organic compounds may be used.



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Material and instruments required

- Pt electrodes
- Calomel reference electrode (accumet)
- mV-pH-Meter (accumet, do not use the yellow meter)
- Cable connecting meter (BNC-connector) and Pt electrode (alligator clip)

Electrode check

The electrodes should be checked in the following solutions:

- Buffered quinhydrone solution: pinch of quinhydrone (it does not dissolve) in ~10 mL of pH 4 buffer (needs to be freshly prepared, not recommendable for field use). Meter should read $+218 \pm 15$ mV.
- Ferrous-Ferric Solution: 39.21g/L Ferrous ammonium sulfate, 48.22 g/L Ferric ammonium sulfate, 56.2 mL/L concentrated Sulfuric acid (recommended for field use since it is reasonably stable compared to buffered quinhydrone). Meter should read $+430 \pm 15$ mV.

1. Connect a calomel reference electrode to mV-pH meter and insert into solution
2. Connect cable fitted with alligator clip and BNC-connector for insertion into the meter.
3. Connect alligator clip to Pt electrode (photo 1) and insert platinum end into solution along with calomel. Record reading when stable (which can take some time).
4. Finally, electrodes should be checked in DI-water. Select electrodes that show a similar reading.

Troubleshooting

1. The platinum must be pure for construction. If not, the electrodes will seem to work fine in a standard solution, but will perform poorly in soil.
2. If none of the electrodes reads correctly in quinhydrone or ferrous-ferric solution, then there is a problem with the calomel, the meter, or the connections. A check of the circuit can be made by connecting the alligator clip to the metal on the meter. This should zero out and give a mV reading of 0. If moving any of the connections causes the mV reading to jump, then the connection is not good.
3. If a few of the electrodes do not read correctly in standard solutions, then the problem lies in those electrodes. The most likely cause is failure to completely remove the oxidation (or epoxy from new electrodes) from the platinum. Remove by scraping lightly with a razor or using wet/dry sandpaper. The section where the alligator clip is attached can be scraped to remove oxidation, but never use the same razor to scrape the platinum. If these steps do not work, then the electrode must be remade. However, make sure that the problem does not lie in the meter, calomel or connections first. The second most likely problem results from failure of the epoxy seal, which allows water to enter and contact the copper wire. This will usually cause anomalously low (negative) readings. If this is the problem, the electrode will have to be dismantled and remade.



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Measurement of Redox Potential in Soil

1. Insert Pt electrode(s) carefully into the soil to the desired depth. Soft soils may require a support to keep electrode upright. The exposed wire or alligator clip should not be immersed or allowed contact with water during measurement. If wetted, they should be dried.

2. Allow electrode to equilibrate for a period of time. This varies depending on the soil and the redox potential (generally, more reducing soils become poised more quickly). A good rule of thumb is 30 minutes to one hour for saline sediments. If several Pt electrodes are available, they can be deployed and allowed to equilibrate while the operator is performing other tasks. Alternatively, measurements of several sites, mesocosms or replicates can be carried out by repeatedly using the same electrodes (at least 3 in order to recognize a malfunctioning electrode).

3. Attach alligator to Pt electrode. Insert calomel electrode into surface water or in contact with wet sediment. The proximity to the platinum electrode does not matter as long as the calomel makes contact with wet soil or surface water to complete the circuit. Do not force the calomel electrode into dry or hard soil.

4. Allow the reading to stabilize and record, sometimes you will have to go with the closest 5 or 10 mV. This can take some time; however, a reading shouldn't take longer than 5 minutes. If several depths of one site/mesocosm are measured, readings should stabilize more quickly with increasing depth. If there are large fluctuations in the reading (i.e., jumping around), check connections and electrode.

5. The readings need to be corrected for the difference between the calomel and the standard hydrogen electrode (SHE). Add the potential of the calomel electrode (+244) to the mV reading. For example, if the measurement is -344 mV, then the calculated Eh is -100 mV. See Table 1 to compare your reading.

Table 1: Thermodynamic sequence for reduction of inorganic substances by hydrogen (at 25°C, pH7)

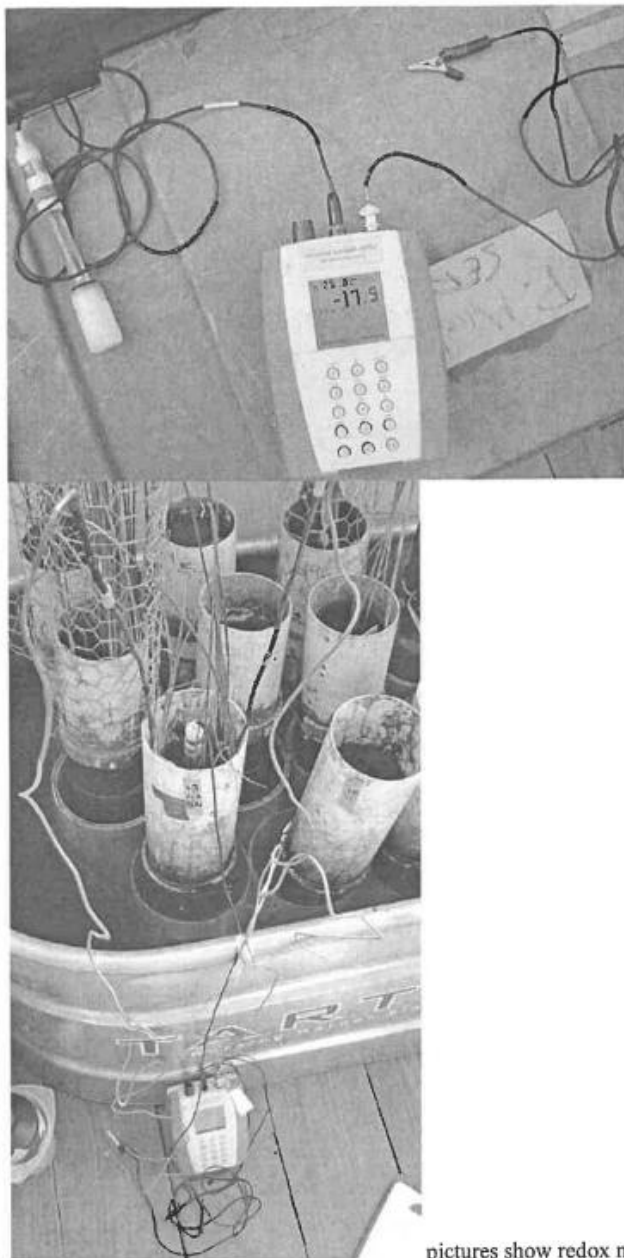
Reaction	Eh (mV)
Reduction of O ₂	812
Reduction of NO ₃ ⁻	747
Reduction of Mn ⁴⁺ to Mn ²⁺	526
Reduction of Fe ³⁺ to Fe ²⁺	- 47
Reduction of SO ₄ ⁻ to H ₂ S	-221
Reduction of CO ₂ to CH ₄	-244

Storage of calomel electrode

Never store electrode in DI water or just dry.

If storage solution provided by manufacturer is empty, you can alternatively prepare a saturated KCl solution by mixing ~ 1g KCl in 200mL pH 7 Buffer.

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pictures show redox measurement on a marsh organ pot.

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C. Supplemental Document
 Example Field Datasheet (link)

COMPASS CB Synoptic Site Sampling Data Sheet										Site Name:	Date:							
										Crew:								
Transect Location	Probe #	Depth (cm):	Reading	Reading	Reading	Reading	Reading	Reading	Reading	Transect Location	Probe #	Depth (cm):	Reading	Reading	Reading	Reading	Reading	Reading
Upland	1									Transition	4							
Upland	1									Transition	4							
Upland	1									Transition	4							
Upland	2									Transition	5							
Upland	2									Transition	5							
Upland	2									Transition	5							
Upland	3									Transition	6							
Upland	3									Transition	6							
Upland	3									Transition	6							
Upland	4									Wetland	1							
Upland	4									Wetland	1							
Upland	4									Wetland	1							
Upland	5									Wetland	2							
Upland	5									Wetland	2							
Upland	5									Wetland	2							
Upland	6									Wetland	3							
Upland	6									Wetland	3							
Upland	6									Wetland	3							
Transition	1									Wetland	4							
Transition	1									Wetland	4							
Transition	1									Wetland	4							
Transition	2									Wetland	5							
Transition	2									Wetland	5							
Transition	2									Wetland	5							
Transition	3									Wetland	6							
Transition	3									Wetland	6							
Transition	3									Wetland	6							

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D. Literature

Megonigal, Patrick, Rabenhorst, Martin. Reduction–Oxidation Potential and Oxygen. (Chapter 5). (2013).

<https://doi.org/10.2136/sssabookser10.c5>

Megonigal, Patrick, Faulkner. Wetland Identification in Seasonally Flooded Forest Soils: Soil Morphology and Redox Dynamics. (1993).

<https://doi.org/10.2136/sssaj1993.03615995005700010027x>

Faulkner, Patrick, Gambrell. Field Techniques for Measuring Wetland Soil Parameters. (1989).

<https://doi.org/10.2136/sssaj1989.03615995005300030042x>

E. Other Photos